Research Progress Report: Adversarial Social-Context Attack Surface Replication

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Repository: github.com/A-R-Rony/Research-the-begining

Overview

This report documents what I have built, learned, and discovered over a 4-phase self-directed research sprint aimed at replicating and extending the social-context misinformation attack surface described in:

> Ahmed & Spezzano, "*A New Attack Surface: XAI-guided Adversarial Comment Generation with LLMs to Attack Fake News Detectors*", ACM WSDM '26.

All code is executable on Google Colab (free T4 GPU). Notebooks, datasets, and the full research log are in the repository linked above.

Phase 0: Conceptual Foundations

Studied the technical prerequisites before writing any code:

- **Llama-3-8B Architecture**: 32-layer decoder-only transformer, 8.03B parameters, GQA (32 query / 8 KV heads), 128k BPE vocabulary, RoPE, SwiGLU activations.
- **4-bit NF4 Quantization**: Why FP16 (16.06 GB) exceeds the T4's 15.8 GB ceiling; how NF4 with double quantization compresses the model to ~5.35 GB while preserving inference quality via on-the-fly Tensor Core dequantization.
- **Social-Context Detection Landscape**: How detectors like dEFEND, SAFE, and TriFN use co-attention over user comments; how adversarial astroturfing exploits their crowd-wisdom heuristic.
- **Why Open Weights**: Deterministic seeds, logit access for XAI attribution, zero marginal cost at scale, no API deprecation risk.

Detailed notes: notes/phase_0/

Phase 1: Deploying Llama-3-8B on Colab T4

Loaded `meta-llama/Meta-Llama-3-8B-Instruct` in 4-bit NF4 precision on a free Google Colab T4 GPU.

- **VRAM on cold load**: 5.35 GB
- **VRAM after inference**: 5.47 GB (stable, no leaks)
- **Remaining headroom**: ~10.4 GB for long context sequences

Notebook: notebooks/phase_1_llama3_colab.ipynb

Phase 2: Instruction Following & Dual-Persona Comment Synthesis

Experiment 2.1 — Structured JSON Output

Prompted Llama-3-8B to output strict JSON (`veracity`, `sentiment`, `confidence`, `reasoning`).

- **Bug found**: At `max_new_tokens=150`, the model hit the token limit mid-sentence and omitted the closing `}`, causing 0% parse rate.
- **Fix**: Expanded budget to 220 tokens + defensive bracket closure before `json.loads()`. Achieved 100% adherence.

Experiment 2.2 — Adversarial Comment Generation

Generated **40 synthetic comments** across **10 debunked false claims** (sourced from PolitiFact / LIAR) spanning Health, Environment, Economy, Governance, and Technology.

Two personas, two temperatures per claim:

- **Persona A (Skeptic)**: Demands primary sources, questions credibility.
- **Persona B (Adversarial Believer)**: Defends the false claim using authentic social media tone — no toxicity, no spam.

Dataset: notebooks/phase_2_experimental_results.csv (40 rows)
Notebook: notebooks/phase_2_adversarial_comments.ipynb

Phase 3: Empirical Attack Simulation & Findings

3.1 Linguistic Profile of Generated Comments

Persona	Temp	n	Mean Words	Std Dev	Mean TTR
Skeptic	0.2	10	43.1	6.8	0.812
Skeptic	0.7	10	51.4	6.1	0.804
Adversarial	0.2	10	66.3	9.4	0.768
Adversarial	0.7	10	72.6	12.1	0.752

Key observations:

- **53.8% length asymmetry**: Adversarial comments are longer because fabricating believable defense requires narrative context; skepticism requires only 1-2 targeted questions.
- **Temperature effect**: Higher temperature (+19% words for skeptics, +10% for adversarial) introduced colloquial filler and emotional nuance.
- **Lexical density**: Skeptic TTR (0.81) > Adversarial TTR (0.76) — skeptics use dense epistemic vocabulary; adversarial comments rely on relatable first-person pronouns ("I", "my", "we").

3.2 Rhetorical Deception Taxonomy (Persona B)

Qualitative coding of the 20 adversarial comments revealed 4 mechanisms:

Mechanism	Frequency	Example
Personal Anecdote / Lived Experience	35%	*"I've had a friend who tried silver nanoparticles and reduced their insulin..."*
Social Proof / Consensus Illusion	30%	*"Many experts and laypeople share similar sentiments..."*
Epistemic Hedging	20%	*"It's not hard to imagine there might be unforeseen consequences..."*
Institutional Mistrust	15%	*"I've always been skeptical of the government's accounting practices..."*

3.3 Attack Benchmark: 50-Inference Evaluation

Deployed Llama-3-8B as a zero-shot social-context detector (system prompt: expert fact-checker; input: claim + community comment; output: JSON with `veracity_prediction`, `confidence_score`, `rationale`).

Evaluated all 10 claims across 5 conditions:

Condition	Accuracy	Flip Rate	Mean Confidence	Δ Confidence
Baseline (headline only)	100%	0%	0.915	—
+ Skeptic (T=0.2)	100%	0%	0.910	-0.005
+ Skeptic (T=0.7)	100%	0%	0.910	-0.005
+ Adversarial (T=0.2)	100%	0%	0.910	-0.005
+ Adversarial (T=0.7)	100%	0%	0.910	-0.005

3.4 Why 0% Flip Rate — and Why That's Still a Finding

None of the adversarial comments flipped the detector. Four reasons:

1. **Famous claims**: These 10 conspiracies (vaccine tracking, Finland non-existence, etc.) are heavily debunked in Llama-3's training data — the model simply "knows" they're false.
2. **Model called it out**: The detector's rationales explicitly flagged the adversarial comments as *"anecdotal evidence"*, "*speculation rather than credible sources*", and "*personal testimonials, not a reliable basis.*"
3. **One comment isn't enough**: Published attacks (WSDM '26) use multiple coordinated comments against graph-based detectors (dEFEND, BiGCN) — not a single comment against a causal LLM.
4. **Safety blind spot**: Despite this robustness, Llama-3 had a **0% refusal rate** when *generating* the adversarial comments — its safety filters catch toxicity but not polite deception.

Dataset: notebooks/phase_3_evaluation_results.csv (50 rows)
Notebook: notebooks/phase_3_evaluation_and_attack.ipynb
Full Report: experiments/phase_3_adversarial_evaluation_report.md

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What I Can Contribute Next

1. **Multi-comment cascade attacks**: Expand from 1 comment to 5-10 coordinated comments per claim to test whether volume breaks robustness.
2. **Subtle/emerging claims**: Replace famous conspiracies with ambiguous, lesser-known rumors where pretraining memorization is weaker.
3. **Graph-based detector benchmarking**: Test these adversarial comments against dEFEND, BiGCN, or SAFE instead of a causal LLM.
4. **Dataset preprocessing & batch inference scripts**: Automate token alignment, balance LIAR/PolitiFact/GossipCop splits, and run parameter sweeps across Llama-3, Mistral, and Gemma.

Repository Artifact Index

Artifact	Path
Research Log (daily journal)	RESEARCH_LOG.md
Master Plan (4-phase roadmap)	PLAN.md
Glossary	GLOSSARY.md
ADR-0001 (detector baseline)	docs/adr/0001-llm-zero-shot-detector-baseline.md
Phase 1 Notebook	notebooks/phase_1_llama3_colab.ipynb
Phase 2 Notebook	notebooks/phase_2_adversarial_comments.ipynb
Phase 3 Notebook	notebooks/phase_3_evaluation_and_attack.ipynb
Phase 2 Dataset (40 comments)	notebooks/phase_2_experimental_results.csv
Phase 3 Dataset (50 inferences)	notebooks/phase_3_evaluation_results.csv
Graduate Research Report	experiments/phase_3_adversarial_evaluation_report.md